
Week 7 · Lecture B

Learning Together

and the Road to Synthetic Populations

Topics: Independent Learners · Cooperative failure · CTDE · LLM-driven policy · GABM preview

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Agenda — 2 hours

30'	Independent Learners: architecture and training loop
30'	Why cooperation fails: non-stationarity as diagnostic
20'	CTDE paradigm — MADDPG as conceptual solution
20'	From reward-driven to goal-driven: the LLM bridge
20'	GABM preview: generative agents and synthetic populations

Where we are coming from: `simple_spread` environment, PettingZoo Parallel API, all from Lecture A. **Where we are going:** synthetic populations as the course finale.

Independent Learners — the baseline approach

Independent Learners (IL): treat every other agent as part of the environment. Each agent i runs its own single-agent RL algorithm (here: PPO) on its local observations. No explicit coordination.

Architecture

For each agent $i = 1, \dots, N$:

- ▶ Observation: o_i^t
- ▶ Actor: $\pi_{\theta_i}(a_i | o_i)$
- ▶ Critic: $V_{\phi_i}(o_i)$
- ▶ Reward: r_i^t
- ▶ Update: PPO

No parameter sharing.

Training loop

```
obs, _ = env.reset()
for step in range(T):
    actions = {a: ppo[a].act(obs[a])
               for a in env.agents}
    obs, rew, term, _, _ = env.step(actions)
    for a in env.agents:
        ppo[a].store(obs[a], actions[a], rew[a])
    if step % T_update == 0:
        for a in env.agents:
            ppo[a].update()
```

Why Independent Learners look appealing

Advantages

- ▶ **Zero engineering overhead:** reuse any single-agent algorithm off the shelf
- ▶ **Scales trivially:** N agents = N independent training runs
- ▶ **Decentralised by design:** no communication, no shared memory
- ▶ **Works well in competitive settings:** self-play, league training (AlphaStar)

The hidden assumption

Each agent treats others as *stationary environment noise*.

But others are learning too.

From agent i 's perspective:

$$P(s' \mid s, a_i) = \sum_{a_{-i}} P(s' \mid s, \mathbf{a}) \pi_{-i}(a_{-i} \mid s)$$

This *effective* transition kernel changes every time any other agent updates its policy.

The environment is non-stationary from every agent's viewpoint — even if the underlying MDP is fixed.

Cooperative failure — non-stationarity as diagnostic

Empirical signature: in `simple_spread` with IL-PPO, agents converge to *suboptimal fixed points* — two agents cluster on the same landmark while the third goes uncovered. Reward variance stays high; convergence is slow or absent.

Three failure modes

1. Lazy agent: reward dominated by others $\Rightarrow$ vanishing gradient.

2. Collision loop: agents chase the same landmark and keep interfering.

3. Credit assignment: shared $r = \sum_i r_i$ hides individual contributions $\Rightarrow$ noisy gradients.

Physics analogy

Each agent minimises local free energy:

$$\mathcal{F}_i = \mathbb{E}_{o_i}[-r_i] + \beta^{-1} H(\pi_i).$$

But r_i depends on π_{-i} , which shifts during training.

Analogy: gradient descent on a potential surface that deforms while being minimised.

Centralized Training, Decentralized Execution

CTDE paradigm: during *training*, allow access to global state and all agents' actions. During *execution*, each agent acts using only its local observation. This breaks the non-stationarity without sacrificing deployability.

MADDPG — key idea

Decentralised actor: $a_i = \mu_{\theta_i}(o_i)$

Centralised critic: $Q_{\phi_i}(x, a_1, \dots, a_N),$

$x = (o_1, \dots, o_N)$

Critic sees joint obs–actions; actor uses only o_i .

At test time: critic discarded.

No comm. needed at execution.

Physics analogy

CTDE $\equiv$ mean field + local execution

- ▶ Critic $\rightarrow$ effective field (global correlations)
- ▶ Actor $\rightarrow$ local field at execution
- ▶ Training: solve coupled system
- ▶ Execution: each spin reads h_i^{eff}

IL vs. CTDE — what changes and what stays

Aspect	Independent Learners	CTDE (MADDPG)	Key gain
Critic input	o_i only	$(o_1, \dots, o_N, a_1, \dots, a_N)$	Stable value estimate
Actor input	o_i only	o_i only	Same deployment
Non-stationarity	Seen by critic	Absorbed by critic	Fixed
Communication	None	None (at exec.)	Deployable
Scalability	$O(N)$	$O(N^2)$ in joint obs	Trades cost
Credit assignment.	Noisy	Per-agent Q_i	Cleaner gradient

Key insight: CTDE does not change the *execution architecture* — only the *training signal*. The deployed agent is identical to an IL agent. The difference is entirely in how the critic was trained.

From reward-driven to goal-driven: the LLM bridge

RL agent

- ▶ Policy: $\pi_{\theta}(a | o)$
- ▶ Behaviour from reward R
- ▶ Needs reward shaping
- ▶ Actions: vectors
- ▶ Memory: hidden state only
- ▶ Communication: via environment

Best for repeated tasks.

LLM agent

- ▶ Policy: $\pi_{\text{LLM}}(\text{action} | \text{context})$
- ▶ Behaviour from language goal
- ▶ No reward shaping
- ▶ Actions: text / tools
- ▶ Memory: retrieved store
- ▶ Communication: language

Best for open-ended tasks.

Bridge: both are agents with state, policy and interaction. RL learns policy from reward; LLMs use pre-trained language models. MARL gives the shared multi-agent structure.

Generative Agents (Park et al., 2023). 25 LLM agents in a sandbox (Smallville) with *memory*, *reflection*, *planning*. Emergent social behaviour from local interactions.

Architecture

Memory: log of observations/actions.

Retrieval: recency + importance + relevance.

Reflection: high-level summaries (e.g. “party”).

Planning: daily schedule updated on-line.

MAS reading

- ▶ Agent = LLM policy node
- ▶ Obs = memory + percepts
- ▶ Action = text → events
- ▶ No reward (persona-driven)
- ▶ Emergence: social patterns
- ▶ Local rules ⇒ global order

Synthetic Populations — GABM at scale

Synthetic population: large MAS where each agent has a *demographic profile* (age, income, health, location) and an LLM-driven behaviour. Used for policy simulation, epidemiology, urban planning.

AgentTorch

[https:](https://agenttorch.github.io/AgentTorch/)

[//agenttorch.github.io/AgentTorch/](https://agenttorch.github.io/AgentTorch/)

- ▶ 10^6+ agents, GPU
- ▶ Profile + LLM policy
- ▶ Apps: COVID, tax, energy
- ▶ Calibrated on real data
- ▶ Counterfactual scenarios

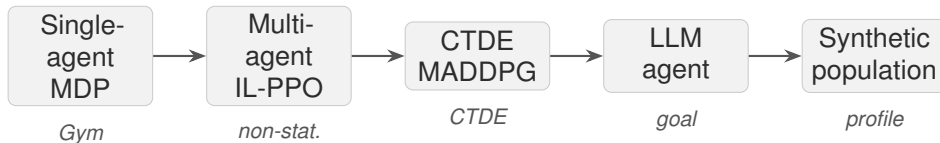
Physics reading

Micro → macro

- ▶ Agents $\equiv$ particles
- ▶ Profile $\equiv$ disorder
- ▶ Policy $\equiv$ interactions
- ▶ Stats $\equiv$ order parameter
- ▶ Phase transitions (epidemics, opinions)

Stat mech with intelligent particles.

The full pipeline: from MDP to synthetic population



What stays

- ▶ Local obs/action agents
- ▶ Emergent collective behaviour
- ▶ Environment abstraction
- ▶ Mean-field diagnostic

What changes

- ▶ $\nabla_{\theta} J \rightarrow$ pre-trained LLM
- ▶ $R \rightarrow$ natural language
- ▶ No memory $\rightarrow$ retrieval
- ▶ Anonymous $\rightarrow$ profiles

Lecture B — Summary

Independent Learners Each agent runs its own PPO on local observations. Simple, scalable, competitive-friendly. Fails in cooperative settings due to non-stationarity.

Cooperative failure Non-stationarity causes lazy agents, collision loops, and credit assignment noise. The reward landscape deforms faster than gradient descent can follow.

CTDE Centralised critic absorbs non-stationarity during training; decentralised actor stays deployable. MADDPG $\equiv$ mean-field training + local execution.

LLM bridge Replace gradient-trained policy with LLM conditioned on persona + context. Goal = natural language, not reward. MARL interaction structure is preserved.

Synthetic populations Large-scale MAS with heterogeneous LLM agents carrying demographic profiles. Statistical mechanics with intelligent, disordered “particles”. Next: your material.

→ **Next:** Generative Agent-Based Models · Synthetic populations · AgentTorch · calibration against real data